# Risk Analysis

**Web-Based Laboratory Inventory and Resource Management System**

Scope: Junior High School Science Laboratory and Technology and Livelihood Education (TLE) Laboratory

## 1. Purpose

This document identifies the major risks that may affect the development, deployment, operation, security, and reliability of the Web-Based Laboratory Inventory and Resource Management System. The analysis focuses on risks related to inventory accuracy, user access, borrowing transactions, data integrity, security, backup and recovery, usability, and system maintenance.

## 2. Risk Rating

Risk level is determined using Likelihood and Impact. Likelihood describes how likely the risk is to occur. Impact describes the possible effect on the system, laboratory operations, data, or accountability.

| Rating | Likelihood | Impact |
|---|---|---|
| Low | Unlikely to occur or limited occurrence | Limited effect and manageable through normal procedures |
| Medium | Possible to occur | May affect operations, data, or users and requires planned controls |
| High | Likely to occur or may occur repeatedly | May significantly affect inventory accuracy, security, availability, or accountability |

## 3. Risk Register

| ID | Risk | Likelihood | Impact | Description | Effect | Mitigation / Control | Owner |
|---|---|---|---|---|---|---|---|
| RA-001 | Incorrect Inventory Data | Medium | High | Inventory quantities, status, condition, or location may not match the actual laboratory resources. | Inaccurate information may affect borrowing decisions, reporting, and accountability. | Use required fields, data validation, controlled inventory adjustments, audit logging, and regular physical inventory checks. | Laboratory Coordinator |
| RA-002 | Unauthorized Access | Medium | High | A user may access information or functions outside their | Unauthorized changes or disclosure of laboratory | Use authentication, server-side role-based authorizatio | System Administrator |

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| | | | | assigned role or laboratory. | information may occur. | n, laboratory access restrictions, secure sessions, and account deactivation. | |
| RA-003 | Weak Password or Credential Security | Medium | High | User credentials may be weak, exposed, or improperly stored. | Unauthorized users may gain access to protected resources. | Use secure password hashing, protected sessions, secure logout, and authorized credential reset procedures. | System Administrator |
| RA-004 | Data Loss | Medium | High | System or database failure may cause loss of inventory and transaction records. | Historical records, inventory information, and borrowing information may become unavailable or permanently lost. | Implement regular database backups, backup retention, secure backup storage, recovery procedures, and periodic backup verification. | System Administrator |
| RA-005 | Inconsistent Inventory and Transaction Data | Medium | High | Related inventory and transaction records may become inconsistent during an operation. | Inventory quantities and borrowing status may become unreliable. | Use database transactions for related updates, data validation, and rules that prevent invalid inventory states. | Development Team / System Administrator |
| RA-006 | Improper Borrowing Release | Medium | High | Items may be released without proper approval or while unavailable. | Resources may be borrowed without authorization or may be assigned incorrectly. | Require approval before release and validate asset availability, status, and | Laboratory Coordinator / Laboratory Staff |

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| | | | | | | condition before release. | |
| RA-007 | Unrecorded Damage or Loss | Medium | High | Damaged or lost resources may not be recorded promptly. | Inventory availability and accountability may become inaccurate. | Provide dedicated damage and loss recording, update resource status, and preserve the related history and audit record. | Laboratory Coordinator / Laboratory Staff |
| RA-008 | Unauthorized Inventory Adjustment | Low | High | An unauthorized user may attempt to change inventory quantities. | Inventory records may be manipulated and accountability may be reduced. | Restrict quantity modification to authorized users, require adjustment reasons, and record changes in the audit trail. | System Administrator |
| RA-009 | Security Vulnerabilities | Medium | High | The web application may be exposed to common vulnerabilities such as SQL injection, XSS, CSRF, broken access control, or session attacks. | User accounts, inventory information, and system operations may be compromised. | Validate and sanitize input, enforce server-side authorization, protect sessions, use secure development practices, and deploy production access through HTTPS. | Development Team |
| RA-010 | Poor User Adoption | Medium | Medium | Users may have difficulty using the system or may continue relying on paper records and | The system may not achieve the expected reduction in manual record keeping. | Use a simple and consistent interface, clear validation messages, understandable terminolog | Project Team / Laboratory Coordinator |

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| | | | | spreadsheets. | | y, and minimize unnecessary steps. | |
| RA-011 | Incorrect User or Laboratory Permissions | Medium | High | A user may be assigned an incorrect role or laboratory access. | The user may view or modify information outside the intended scope. | Validate account setup, restrict laboratory access by assignment, and allow only authorized administrators to manage permissions. | System Administrator |
| RA-012 | System Downtime | Medium | Medium | The web application or its supporting services may become temporarily unavailable. | Users may be unable to access inventory or process borrowing and return transactions. | Maintain backup and recovery procedures, monitor system availability, and provide recovery procedures for service failures. | System Administrator |
| RA-013 | Incomplete Audit Trail | Low | High | A significant inventory or security-sensitive activity may not be logged correctly. | Accountability and traceability may be reduced. | Log significant activities with timestamp, user, action, affected entity, record ID, previous value, new value, and reason where applicable. | Development Team |
| RA-014 | Incorrect Return Processing | Medium | High | Returned quantity, missing quantity, damaged quantity, or condition may be recorded | Inventory and asset status may not reflect the actual returned resources. | Validate that return quantity does not exceed released quantity and require return | Laboratory Staff |

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| | | | | incorrectly. | | condition and related information. | |
| RA-015 | Maintenance Status Not Updated | Medium | Medium | An asset may remain available or borrowable while it is under maintenance. | A damaged or unavailable asset may be released, affecting safety, operations, and inventory accuracy. | Maintain status separately from condition and prevent release of assets under maintenance. | Laboratory Coordinator / Laboratory Staff |
| RA-016 | Future Feature Overreach | Low | Medium | Future intelligent features may be treated as required parts of the initial system. | Development may become more complex and core inventory functions may be delayed. | Keep AI, machine learning, predictive analytics, RFID, barcode or QR scanning, mobile application, and other listed future features outside the initial scope. | Project Team |

## 4. Risk Priorities

The highest priority risks are those that may directly affect inventory accuracy, unauthorized access, data loss, transaction integrity, borrowing control, and accountability. These risks shall receive controls during system design and implementation.

| Priority Area | Related Risks | Required Focus |
|---|---|---|
| Inventory Accuracy | RA-001, RA-005, RA-008, RA-014 | Validation, controlled updates, transaction consistency, and audit logging |
| Security and Access | RA-002, RA-003, RA-009, RA-011 | Authentication, server-side authorization, secure input handling, and permission control |
| Data Protection | RA-004, RA-012 | Regular backups, recovery procedures, secure storage, and service recovery |
| Borrowing and Resource Control | RA-006, RA-007, RA-014, RA-015 | Approval, release validation, return validation, and status management |
| Accountability | RA-008, RA-013 | Audit trail and controlled inventory changes |
| Usability and Adoption | RA-010 | Simple interface, clear validation, |

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| | | and efficient common operations |
| Scope Control | RA-016 | Maintain the defined initial scope and separate future enhancements |

## 5. Risk Monitoring

- Risks shall be reviewed during requirements analysis, system design, development, testing, deployment, and evaluation.
- A risk may be re-rated when its likelihood or impact changes.
- High-impact risks shall have defined controls before the affected function is considered ready for use.
- Security, inventory accuracy, data integrity, backup, and auditability shall be included in system testing.
- Issues found during testing or actual laboratory use shall be recorded and assessed for their effect on existing risks.
- Risk controls shall be reviewed together with the related functional, security, data integrity, and backup requirements.

## 6. Risk Acceptance

The project shall prioritize risks that can affect inventory accuracy, accountability, security, and availability. Risks related to future enhancements shall not be allowed to delay the core inventory and resource management functions. The system shall remain fully functional without the future intelligent capabilities identified in the SRS.

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